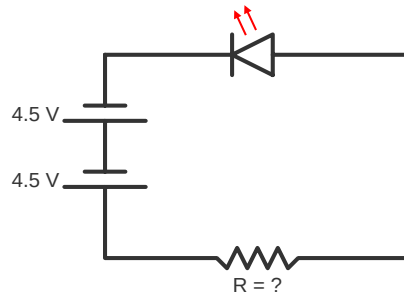


1. A blue LED has a forward voltage of 3.2 V and should run at 20 mA. The circuit uses two 4.5 V batteries as shown. What resistance is needed?



2. Why does an LED need to be placed in the correct direction in a circuit?

3. A small red indicator LED has a 2.0 V drop and is connected to a 5.0 V supply through a 150 Ω resistor. What current flows through the LED?

4. Why can thermal runaway be a problem for an LED?

5. A green LED has a 2.2 V drop and should run at 15 mA from a 6.0 V battery pack. What resistor should be placed in series with it?

6. A red indicator LED has a 2.0 V drop and runs at 20 mA from a 5.0 V supply through a 150 Ω resistor. How much power does the LED use, and how much power turns into heat in the resistor?

7. Two red LEDs are wired in series with one resistor and a 9.0 V battery. Each LED has a 2.0 V drop and should run at 20 mA. What resistor is needed?

8. Why is a transistor useful as a switch in a logic circuit?

9. A NOT gate uses 5.0 V for high and 0 V for low. If the input is 5.0 V, what output voltage should it produce?

10. An AND gate uses 5.0 V for high and 0 V for low. Input A is 5.0 V and input B is 0 V. What output voltage should it produce?

11. An LED circuit has a 9.0 V battery and should use 20 mA of current. What resistor should be used?

12. A student builds a circuit with a resistor. A multimeter set to measure voltage, connected across the resistor, reads 3.3 V. A multimeter set to measure current, connected in series with the resistor, reads 15 mA. What is the resistance?

13. An OR gate uses 5.0 V for high and 0 V for low. Input A is 0 V and input B is 5.0 V. What output voltage should it produce?

14. An XOR gate uses 5.0 V for high and 0 V for low. If both inputs are 5.0 V, what output voltage should it produce?